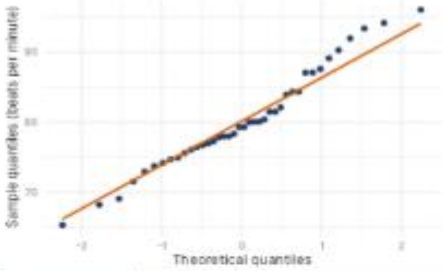
DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot D

R.No: 29 Reg. No: 192524136 Name: Peram Mahitha

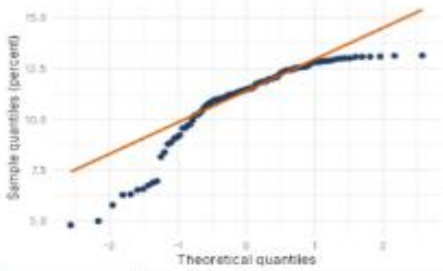
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: resting pulse (beats per minute), n = 40



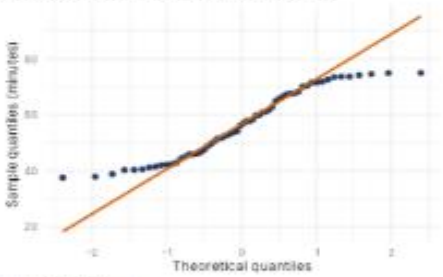
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: battery drain per hour (percent), n = 100



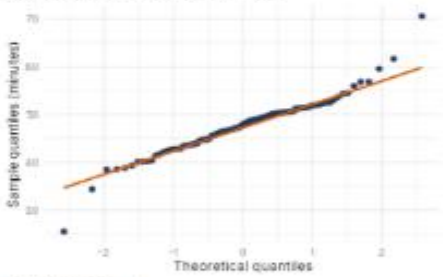
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: lab experiment duration (minutes), n = 60



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: commute time (minutes), n = 100



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 4 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Chart 1: Resting Pulse (beats per minute), n = 40

1. Normal (Yes / No): No

2. Shape (from the line): Approximately normal distribution.

3. Evidence from the plot: Most of the points lie close to the reference line, showing that the resting pulse values follow an approximately normal distribution. However, a few points deviate from the line at the upper end, indicating slight differences from a perfect normal distribution.

Chart 2: Battery Drain per Hour (percent), n = 100

1. Normal (Yes / No): No

2. Shape (from the line): Left-skewed distribution.

3. Evidence from the plot: The points deviate from the reference line at the lower end and form a curved pattern. This indicates that the data are not normally distributed and that the lower battery-drain values create a longer tail on the left side.

Chart 3: Job Applicants' Duration (days), n = 60

1. Normal (Yes / No): No

2. Shape (from the line): Right-skewed distribution.

3. Evidence from the plot: Most points follow the reference line in the middle, but the points deviate at the lower and upper ends. The upper-end deviation suggests a longer right tail, indicating that a few applicants have much longer durations than the others.

Chart 4: Commute Time (minutes), n = 100

1. Normal (Yes / No): No

2. Shape (from the line): Slightly right-skewed distribution.

3. Evidence from the plot: Most points lie close to the reference line, indicating that the commute times are approximately normal in the middle. However, some points at the upper end lie above the line, suggesting a few unusually long commute times and a slight right-skewed pattern.

Part B: R Code (4 Marks)

B1. Chart 4 shows a slight departure from normality because a few points at the upper end deviate from the reference line. I would report the median and interquartile range (IQR) along with the mean and standard deviation, as these measures are less affected by unusually long commute times.

B2. To draw a Q-Q plot with a reference line in R for a data frame column named commute_time, use:

```
qqnorm(df$commute_time,
    main = "Q-Q Plot of Commute Time")
qqline(df$commute_time, col = "red")
```